KAMSSA UACE PHYSICS PAPER 1
SCORING GUIDE 2026

ITEM 1

At the cathode electrical energy gained is converted into kinetic energy.

$$eV_T = \frac{1}{2}mu^2$$

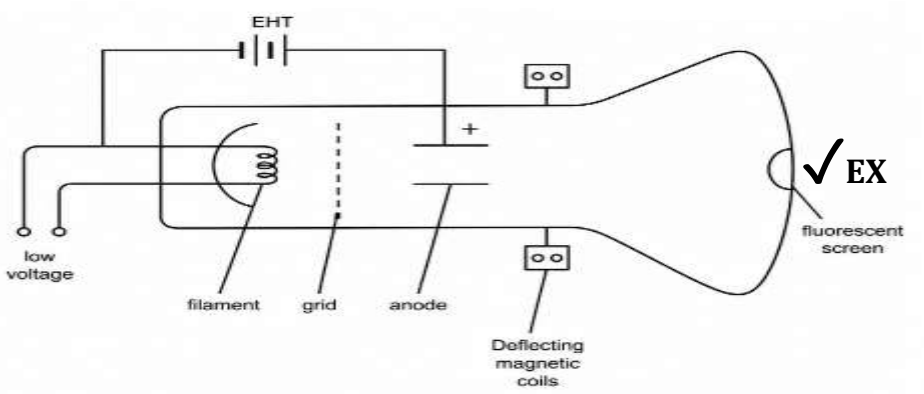
$$u = \sqrt{\frac{2\times7.7\times1000\times1.6\times10^{-19}}{9.11\times10^{-31}}}$$ ✓**AN**

$u = \mathbf{5.2\times 10^{7} ms^{-1}}$ ✓**AN** **03 scores**

The screen gives normal brightness if electron strikes the screen with speed of 5.2X10 this condition is satisfied thus the screen gives normal brightness ✓**AN**

Production of visible image by a C.R.T television

Figure 1;a Cathode Ray Tube



✓**EX** **05 scores**

- The cathode filament is heated and releases electrons by thermionic emission. ✓**EX**
- The positive anode accelerates the electrons towards the screen.
- The control grid controls the number of electrons accelerated to the screen and controls brightness. ✓**EX**
- Magnetic coils located on the outside of the tube neck generate magnetic fields that bend the electron beam to the extract position on the screen.
- The inside of the fluorescent screen is coated with phosphorous material that glows when hit by electrons. ✓**EX**
- The beam sweeps across the screen horizontally and vertically at high frequency creating the image. ✓**EX**

Brightness control;

When light falls on the photo sensitive metal plate electrons are emitted from the metal surface by photo electric effect. The emitted electrons produce a current that is sent to the control grid. In bright light, more electrons are emitted, photocurrent increases, the grid voltage increases and more electrons are directed to the screen and brightness increases. ✓**EX** ✓**EX** ✓**EX** ✓**EX** ✓**EX** ✓**EX** **06 scores**

In weak light; ✓**EX** ✓**EX**

− Few photo electrons reduce, grid voltage reduces, less electrons reach the screen and it becomes dimmer. ✓**EX** **03 scores**

$Brightness\ remaining\ the\ same\ in\ sharp-\ orange\ light;$

$$Photon\ energy;\quad E = ht = 6.63\times10^{-34}2.9\times10^{14}$$

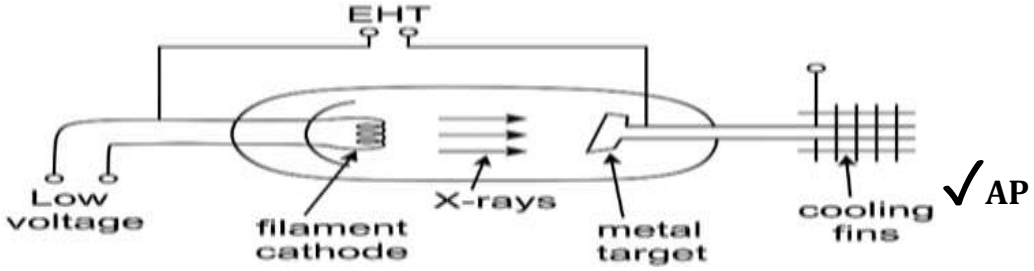
$$= 1.9227\times10^{-19}J$$ ✓**AR**

$$E = \frac{1.9227\times10^{-19}}{1.6\times10^{-19}} = \mathbf{1.20\ eV}\ \checkmark\mathbf{AN}$$

03 scores

Since work function $W_0 > 1.20eV$, electrons are not emitted, no photocurrent flows to the grid and the brightness on the screen remains the same. ✓**AN**

b) *Device; X-ray tube*



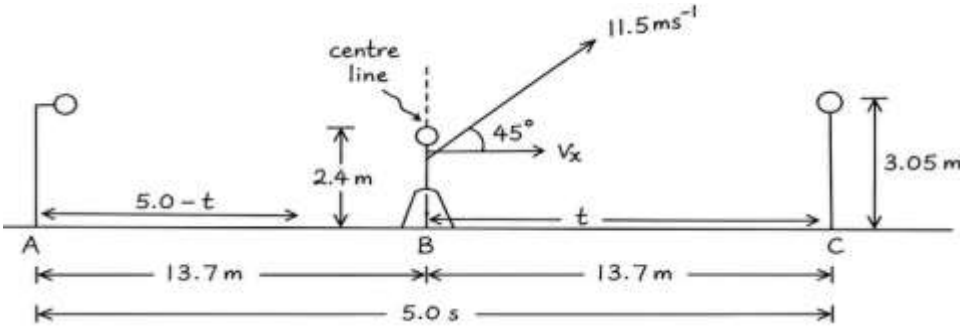
✓**AP**

Figure 2;an X-ray tube

05 scores

- Low voltage heats filament, electrons are emitted thermionically. ✓**AP**
- Emitted electrons are accelerated by a high voltage towards the metal target ✓**AP**
- On striking the metal target, 99% of their kinetic energy is converted into heat an into X-rays and excess heat generated at the target is conducted by copper anode to cooling fins for continuous production of X- rays ✓**AP** ✓**AP**

ITEM 2



Time of projecting the ball;

$(11.5\ cos\ 45^0)t = 13.7, t = 1.685s$ ✓✓

Time taken by the player to reach the centre line;

$$T = 5.0 - 1.685 =$$
$$= \mathbf{3.315}s\ \checkmark \qquad \textbf{AN - 04}$$

The ball was projected $3.315s$ after starting to splint since $3.315s$ ✓< $3.50s$. The ball was projected before the buzzer. Thus, the score counted / valid

Motion of the player

Uniform acceleration; $u = 0, s = 13.7m, t = 3.315s$

$$S = ut + \frac{1}{2}at^2$$

$$13.7 = 0 + \frac{1}{2}a(3.315)^2, \checkmark\mathbf{AN} \qquad a = 2.493ms - \mathbf{2}\ \checkmark\mathbf{AN} \qquad \textbf{AN = 05}$$

Final velocity; $V = u + at$

$V = 0 + 2.493 \times 3.315$ ✓

$$V = \mathbf{8.264ms^{-1}}$$ ✓

The player starts from rest, accelerates uniformly at $2.49ms^{-2}$ ✓ and reaches a velocity of $8.264ms^{-1}$ just before releasing the ball.

Motion of the ball;

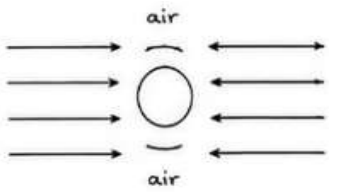
- The ball moves in a parabolic path. ✓**EX**
- The ball moves with constant horizontal velocity and uniform vertical acceleration due to gravity exerting a constant vertical force mg constantly pulling the ball vertically downwards. This makes the ball move along a parabolic path. ✓**EX** ✓**EX** **03 scores**

b) ***David closing remarks;***

Jump shot. **AP - 04**

- The player holds a stationary position. The players horizontal velocity is zero, the ball is then projected only with initial velocity, if a player has a horizontal velocity the actual velocity of the ball would be the sum of the players velocity and the velocity projection of the ball. ✓ ✓ ✓
- This would increase the horizontal velocity of the ball and accuracy would reduce. ✓
- A follow through; helps to provide extended time during throwing. **AP - 04**
- The ball gets enough required momentum in a controlled way. ✓ ✓ *From impulse = Force X time*.
- The ball attains increased velocity with minimum force and velocity is in a particular intended direction. ✓ ✓

Snap shot creates a back spin



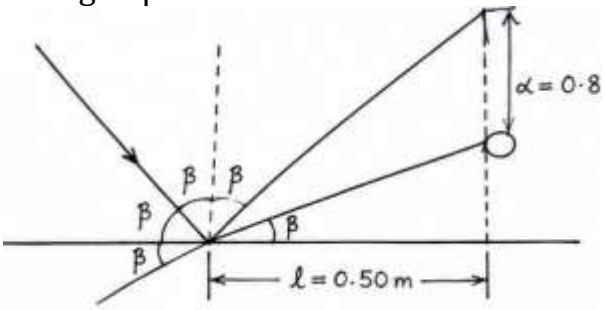
✓**AP**

AP - 05

- As the ball spins in air, there is increased velocity of air above the ball, the pressure there reduces. The reduced air velocity below the ball then increases the pressure. ✓
- The ball experiences a small uplift due to this pressure difference. The ball keeps in air longer ✓ and moves at a controlled speed. This improves the accuracy of the target.

ITEM 3

When the reflecting water surface is tilted through a small angle β. The reflected ray turns through 2β.



small angles in radians;

$$\tan 2\beta \approx 2\beta = \frac{x}{l}$$ ✓

$$\beta = \frac{0.8}{2\times5} = 0.08\ radians$$ **EX - 03**

$$Wave\ number\ K = \frac{2\pi}{\lambda} = \frac{2\pi}{20} = \frac{\pi}{10}\ radian\ m^{-1}$$

$$Amplitude, A = \frac{\beta}{K} = \frac{0.08}{\frac{\pi}{10}} = 0.255m$$

The device can generate power if $A > 0.25m$. Since $A = 0.255m > 0.25m$, the project is supported by the lake waves. EX - 1

$$T = 5.0s, f = \frac{1}{T} = \frac{1}{50} = 0.20Hz$$ **AN - 04**

$$\textbf{\textit{Using}};\ y = A\sin(2\pi ft - kx)$$

$$y = 0.255\sin(2\pi \times 0.2t - \frac{\pi}{10}x$$

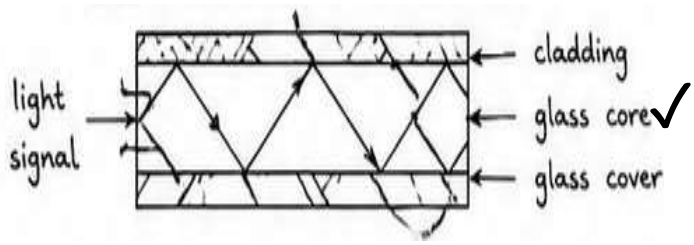
$$y = 0.255\sin(0.4\pi t - \frac{\pi}{10}x)$$

b) *convex lens uses;*

− The reflected beam from the wave may spread. The convex lens collects the light and converges it. EX - 04

−The beam is then focused into the small opening of the optical fiber. Almost all the reflected light is focused into the optical fibre. This improves signal strength.

Optical fibre signal transmission



− Optical fibre is made of a cove that has more refractive index than cladding. EX - 05

−Light is directed at the boundary the glass cove and the cladding at an angle greater than the critical angle

−Light is totally internally reflected, the same happens inside the tube at points of incidence, multiple total internal reflections help to transmit light through the optical fibre without loss of intensity. This method is accurate because it reduces human errors and improves sensitivity.

ITEM4

a) Properties of the radiation;

−Can travel through vacuum.

−Moves at a speed of $3 \times 10^8 ms^{-1}$ in vacuum. **05 scores**

−Has a shorter wavelength / higher frequency

−Has a shorter wavelength electric and magnetic fields

Locating focus point;

−A concave mirror reflects parallel rays onto the focal point.

−The concave mirror is made to face directly towards the sun. **05 scores**

−The blackened bulb of the thermometer is moved along the principal axis of the mirror.

−The position is adjusted until it shows the maximum temperature reading, the bulb absorbs radians reflected from the mirror much at this position and it's the focal point.

Working process of the system.

−Electromagnetic waves from the sun are reflected by the concave mirror onto the focus where all the intensity will be concentrated.

−The energy is absorbed by the blackened bulb of the thermometer.

−The thermometer converts thermal energy into electric pulses which enables the digital thermometer to read the temperature, if the temperature is high enough, it implies that the heat concentrated at that point can be used for cooking. **04 scores**

Maximum temperature;

$$Power\ received\ by\ the\ mirror = Power\ absorbed\ by\ the\ bulb\ of\ the\ thermometer$$

$$\mathrm{I}\,A = 4\pi r_b^2 \sigma T_b^4$$

$$100\pi(0.1)^2 = 4\pi(5\times10^{-3})^2 \times 5.67\times10^{-8}T_b^4$$

$$T_b = 648.044k, \quad T = 648.044 - 273$$

$$T = 375.044℃$$

$$Energy\ absorbed\ by\ the\ bulb = C(\theta_2 - \theta_1)$$

$$pt = 70(375.044 - 20)$$

$$100\pi(0.1)^2 t = 24853.08$$

$$t = 7909.96s = \frac{7909.96}{60}$$

06 scores

$$\boldsymbol{t = 131.83\ minutes}$$

Since $131.83\ minutes > 105\ minutes$, the system is not effective for cooking.

Practical improvements

−Increase the diameter of the concave mirror

−Using a parabolic mirror to increase the radiation intensity at the focus.

−Silvering the inner walls of the container.

−Lagging parts of the cooking utensils leaving only the exposed parts to the radiation

04 scores

b) ***Solar energy utilization*;**

−The effect is applied in solar panels which absorbs electromagnetic radiation / light to produce photo electric current. The photoelectric current charges the battery which is then used to produce electricity for used. **03 scores**

ITEM 5

a) **1st test:** $V_R = Kl$

Case 1; $15\mathrm{I} = 779K \longrightarrow (1)$

$$E = \mathrm{I}(r + R)$$

$$E = \frac{779K}{15}(r + 15 + 27) \longrightarrow (2)$$

Case 2; $15\mathrm{I}_1 = 656K \longrightarrow (3)$

$$E = \frac{656}{15}(r + 15 + 35) \longrightarrow (4)$$

07 scores

Equating eqn (2) ***and*** (4)

$$\frac{779K}{15}(r + 42) = \frac{656K}{15}(r + 50)$$

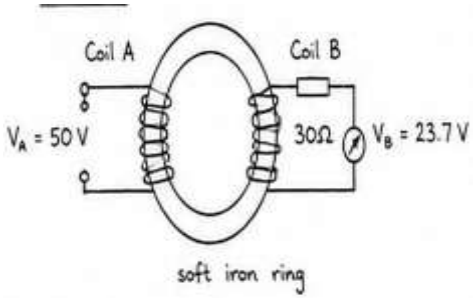
$$\frac{779}{656}(r + 42) = r + 50$$

$1.1875r + 49.875 = r + 50$ ✓

$0.1875r = 0.125$ ✓

$r = 0.67\Omega$

2nd test: $V_B = IR = 30 \times 0.79 = 23.7V$: ✓



04 scores

$\frac{V_A}{V_B} = \frac{N_P}{N_S}, \frac{E}{23.7} = \frac{N_P}{3N_P}$ ✓

$\boldsymbol{E = 7.9V}$ ✓

Since $r < 0.9\Omega$ and emf lies in the range, then the battery is of good quality and customers' complaints were not valid. ✓ ✓

Explanation of heat;

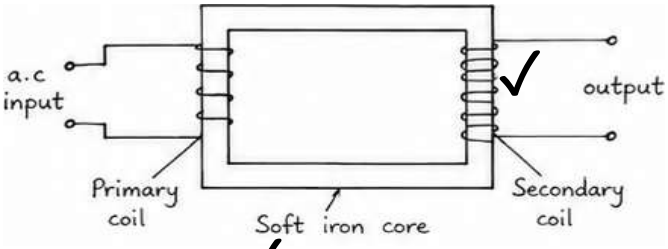
Moving electrons pass through the resistor with much energy when they collide with an atom vibrating about its mean position, they give up some of their kinetic energy to the atom. This increases the amplitude of vibration of the atoms due to collisions which generates heat in a realistic operation. ✓ ✓ ✓ ✓

05 scores

c) The device which uses the setup is a transformer ✓

—It is applied in power stations to step up a voltage. This helps in generating high voltages into transmission lines. ✓ ✓

How it works



09 scores

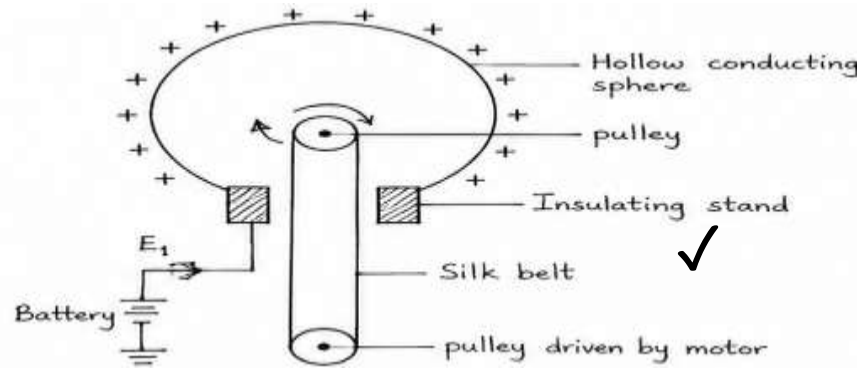
—A.C voltage is fed in the primary coil. ✓

—Alternating current in the primary coil produces an alternating flux in the soft iron core which links up to the secondary coil which produces induced emf in the coil. ✓

—Since the number of turns in secondary coil is greater than those in the primary coil, the voltage is stepped in the secondary coil since emf is proportional to the number of turns. ✓

ITEM 6;

<u>Working of van de Graaff generator</u>



02 scores

−Electrode E_1 is made positive relative to earth, at the E_1 there is high charge density and high electric field intensity, air around E_1 gets ionized

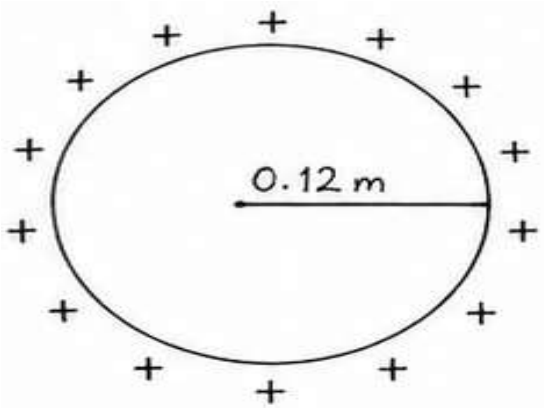
−The positive ions are repelled to the belt which drives them up to the sphere. At the sphere, negative charge is induced at E_2 and positive charge induced at the surface of the sphere.

−Point action takes place at E_2 negative ions are repelled to the belt and discharges it before it moves over the pulley,

−The process continues until the charge on the sphere can ionize the air around it.

05 scores

Energy stored by the sphere;



$Q = 2.4 \times 10^{-6} C$

***Capacitance*;**

$C = 4\pi\varepsilon_0 r$

$C = 4\pi(8.85 \times 10^{-12}) \times 0.12 = 1.33 \times 10^{-11} F$

$E = \frac{1}{2}\frac{Q^2}{C} = \frac{(2.4\times10^{-6})^2}{2\times1.33\times10^{-11}} = 0.216J$

06 scores

Energy supplied by the motor ;

$E = 1Vt = 0.5 \times 12 \times 60 = 360J$

$Efficiency\ of\ conversion\ of\ energy$;

$\eta = \frac{0.216}{360} \times 100\% = 0.06\%$

$Required\ efficiency\ of\ conversion\ of\ energy\ is\ 0.05\% < 0.06\%$

- For the van de Graaff generator to be acceptable, the sphere must be smooth, spherical and clean, thus charge must be uniformly distributed over the surface. To check for this;

- A proof plane is placed on the surface of the dome that is charged and then transferred into a metal can placed on the cap of a neutral gold leaf electroscope.
- Divergence on the gold leaf electroscope is noted
- The experiment is repeated with the proof plane placed at different parts of the dome and corresponding diverges noted.
- If divergencies are equal, the charge is then uniformly distributed.
- Since efficiency of energy conversion is greater than 0.05%. If the charge is uniformly distributed over the dome, the generator can be accepted. **07 scores**

b) As the van de Graaff is kept running charge on the dome kept on increasing until a value when the insulation of the air breaks down.

$$Since\ E = \frac{Q}{4\pi\varepsilon_0 r^2}, Q = (3.0 \times 10^6) \times 4\pi\varepsilon_0 r^2$$

$$Q = (3.0 \times 10^6)(24\pi \times 8.85 \times 10^{-12})(0.12)^2 \quad \textbf{AN - 03}$$

$$\boldsymbol{Q = 4.8 \times 10^{-6} C}$$

When the charge increases to $4.8 \times 10^{-6} C$, the electric field intensity is high enough to break down insulation of the air around the sphere, thus creating sparks around the dome. **EX - 02**

END